

Statistical Programming Languages

Basic statistics & exploratory data analysis

Katarzyna Reluga¹

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Basic terms I

- ▶ **Statistical unit/instance/record**
 - ▶ A single entity being studied, e.g. a person, a household, a company, a country, ...
- ▶ **Variable/feature/predictor**
 - ▶ An attribute measured on each instance
 - ▶ ML also distinguishes the **label/target** = outcome variable to predict.
- ▶ **Observation/sample/row/feature vector**
 - ▶ All feature values (and possibly the label) for one instance.
- ▶ **Data set/matrix**
 - ▶ A set of observations and variables (`data.frame`)
 - ▶ Observation: row in data matrix, variable: column in data matrix
- ▶ **Population**
 - ▶ The full set of entities from which data could be drawn; membership is well-defined.
 - ▶ For each entity we must be able to decide if it belongs to the population or not
- ▶ **Sample:**
 - ▶ A subset of entities from the population
 - ▶ In ML, the word *sample* is often used to mean a **single observation**; in stats it means a **subset**. Be mindful of context.

Basic terms II

```
# Iris flower data set (see Wikipedia)
head(iris, 3)
##      Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1           3.5           1.4           0.2  setosa
## 2           4.9           3.0           1.4           0.2  setosa
## 3           4.7           3.2           1.3           0.2  setosa

# Observations: all census districts (suburbs) of Boston (population)
data(Boston, package = "MASS");dim(Boston)
## [1] 506  14
Boston[c(1:2), c(1:5)]
##      crim zn indus chas    nox
## 1 0.00632 18  2.31    0 0.538
## 2 0.02731  0  7.07    0 0.469
# draw a sample with n = 10 from the population
set.seed(1728)
index <- sample(nrow(Boston), 10)
index
## [1] 381 332 127  35  29 448 217 182 271 123
dim(Boston[index, ])
## [1] 10 14
```

Level of measurement (scale of measure)

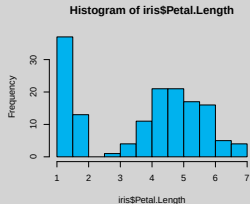
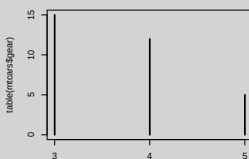
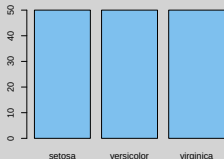
- ▶ a classification of variables
- ▶ describes the nature of information within the values assigned
- ▶ determines which methods and graphics can be applied
- ▶ levels (ascending order of information contents)
 - ▶ **nominal** (logical, factor, character): values can be compared ($=$, $\neq$)
 - ▶ **ordinal** (ordered): values can be sorted ($<$, $\leq$, $=$, $\neq$, $\geq$, $>$)
 - ▶ **metric** : values can be ordered and math operations can be used ($+$, $-$, $/$, $*$)
 - ▶ **continuous** (numeric)
 - ▶ **discrete** (numeric, integer)

```
sapply(iris, class)
## Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## "numeric" "numeric" "numeric" "numeric" "factor"
```

Distribution (CDF/PDF/PMF)

- ▶ It is a building block in statistics/data analysis/ML
- ▶ It describes values a variable can take in the population and its
 - ▶ frequency (variable),
 - ▶ probability/density (discrete/continuous random variable), or
 - ▶ cumulative distribution functions (at least ordinal)

```
par(mfrow = c(1, 3), bg = "lightgrey")  
barplot(table(iris$Species), col = "skyblue2") # nominal  
plot(table(mtcars$gear))                       # metric discrete  
hist(iris$Petal.Length, col = "deepskyblue2") # metric continuous
```

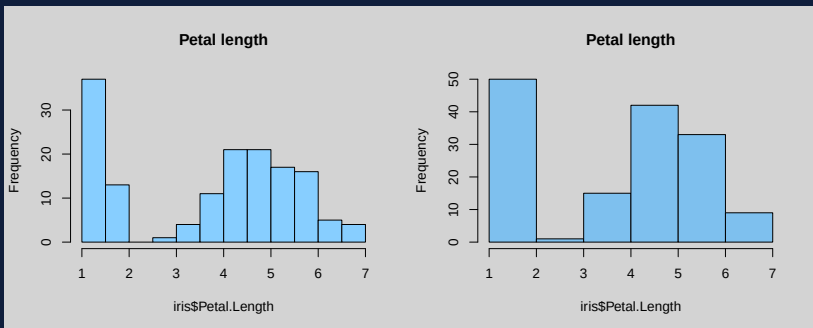


Tip: For more colours, see the [R colours cheat sheet](#).

Histogram I

- We can customize the choice of number of bins.

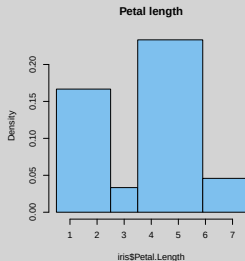
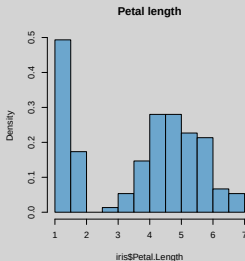
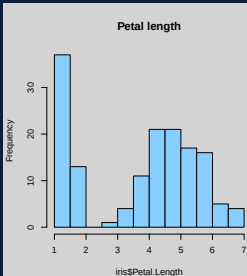
```
par(mfrow = c(1, 2), bg = "lightgrey")  
hist(iris$Petal.Length, breaks = 20,  
     main = "Petal length", col = "skyblue1") # approx. 20 bins R uses  
                                                # pretty() to choose edges  
hist(iris$Petal.Length, breaks = 1:7,  
     main = "Petal length", col = "skyblue2") # exact bin edges
```



Histogram II

- If equidistant bins then the histogram area sums to n otherwise 1 (or choose `freq=FALSE`)

```
par(mfrow = c(1, 3), bg = "lightgrey")
hist(iris$Petal.Length, main = "Petal length", col = "skyblue1")
h <- hist(iris$Petal.Length, freq = FALSE,
          main = "Petal length", col = "skyblue3")
sum(h$density * diff(h$breaks)) # 1
## [1] 1
hist(iris$Petal.Length,
     breaks = c(0.5, 2.5, 3.5, 5.9, 7.5),
     main = "Petal length", col = "skyblue2")
```

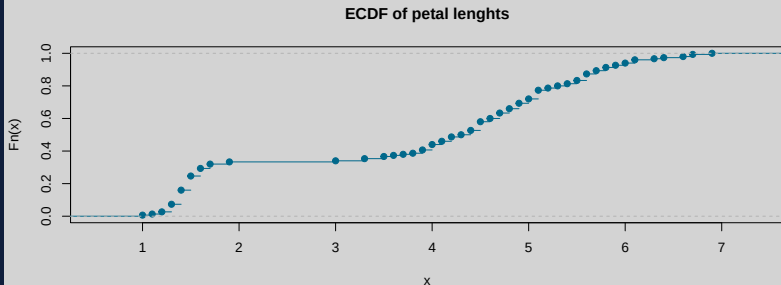


Frequency tables & frequency density

```
table_iris <- table(iris$Species)           # absolute frequencies
table_iris
##
##      setosa versicolor  virginica
##      50      50      50
prop.table(table_iris)                      # relative frequencies
##
##      setosa versicolor  virginica
## 0.3333333 0.3333333 0.3333333
res <- hist(iris$Petal.Length, plot = FALSE) # frequency density
str(res) #str(): displays the internal structure of R objects
## List of 6
## $ breaks   : num [1:13] 1 1.5 2 2.5 3 3.5 4 4.5 5 5.5 ...
## $ counts   : int [1:12] 37 13 0 1 4 11 21 21 17 16 ...
## $ density  : num [1:12] 0.4933 0.1733 0 0.0133 0.0533 ...
## $ mids     : num [1:12] 1.25 1.75 2.25 2.75 3.25 3.75 4.25 4.75 5.25
## $ xname    : chr "iris$Petal.Length"
## $ equidist: logi TRUE
## - attr(*, "class")= chr "histogram"
```

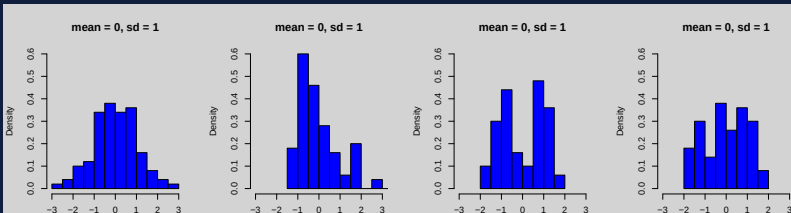

Empirical cdf

```
par(bg = "lightgrey")
plcdf <- ecdf(iris$Petal.Length)
str(plcdf) # 3 classes:
## function (v)
## - attr(*, "class")= chr [1:3] "ecdf" "stepfun" "function"
## - attr(*, "call")= language ecdf(iris$Petal.Length)
plcdf(5) # evaluates the CDF at 5 (the proportion of petals <= 5)
## [1] 0.72
plot(plcdf, main = "ECDF of petal lenghts", col = "deepskyblue4")
```



Describing a distribution (I): location & spread

- ▶ purpose: to summarise a distribution to a few numbers or values
- ▶ location parameter
 - ▶ nominal: mode (the most frequent value)
 - ▶ ordinal: mode, quantiles, median
 - ▶ metric: mean, median, min, max, quantiles
- ▶ dispersion parameters
 - ▶ nominal, ordinal: –
 - ▶ metric: standard deviation, variance, range, interquartile range, variation coefficient
- ▶ loss of information: different distributions may have the same parameters



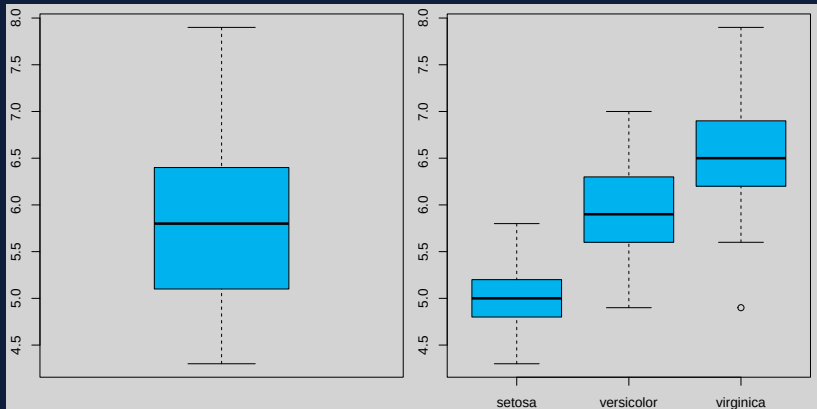
Describing a distribution (II)

```
# location
mean(iris$Sepal.Length)      # mean, non-robust
## [1] 5.843333
median(iris$Sepal.Length)    # median, robust
## [1] 5.8
quantile(iris$Sepal.Length)  # quantile, robust
##   0%   25%   50%   75%  100%
##  4.3   5.1   5.8   6.4   7.9
quantile(iris$Sepal.Length, c(0.1, 0.9))
## 10% 90%
## 4.8 6.9
# dispersion
sd(iris$Sepal.Length)        # standard deviation, non-robust
## [1] 0.8280661
var(iris$Sepal.Length)       # variance, non-robust
## [1] 0.6856935
IQR(iris$Sepal.Length)       # interquartile range, robust
## [1] 1.3
```

- ▶ Remark: `mode` and `range`: not implemented in base R, but in packages
- ▶ Reminder: to remove missing values set the parameter `na.rm = TRUE`

Summarize distribution parameters I

```
par(mfrow = c(1,2), mar = c(2, 2, 0.5, 0.5) + 0.1, bg = "lightgrey")
boxplot(iris$Sepal.Length, col = "deepskyblue2")
boxplot(Sepal.Length~Species, data = iris, col = "deepskyblue2")
```



Summarize distribution parameters II

```
summary(iris)
##      Sepal.Length      Sepal.Width      Petal.Length      Petal.Width
##  Min.      :4.300    Min.      :2.000    Min.      :1.000    Min.      :0.100
##  1st Qu.:5.100    1st Qu.:2.800    1st Qu.:1.600    1st Qu.:0.300
##  Median :5.800    Median :3.000    Median :4.350    Median :1.300
##  Mean   :5.843    Mean   :3.057    Mean   :3.758    Mean   :1.199
##  3rd Qu.:6.400    3rd Qu.:3.300    3rd Qu.:5.100    3rd Qu.:1.800
##  Max.   :7.900    Max.   :4.400    Max.   :6.900    Max.   :2.500
##           Species
##  setosa      :50
##  versicolor:50
##  virginica  :50
##
##
##
```

Summarize distribution parameters III

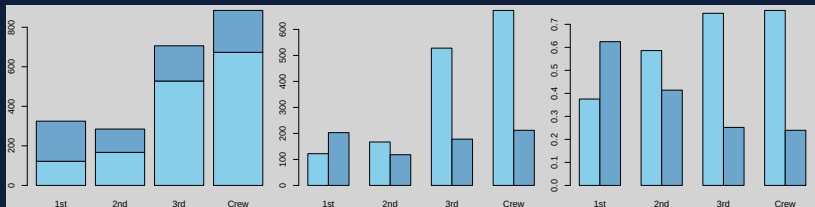
```
psych::describe(iris)
##          vars  n mean   sd median trimmed  mad min max range skew
## Sepal.Length  1 150 5.84 0.83   5.80   5.81 1.04 4.3 7.9   3.6 0.31
## Sepal.Width   2 150 3.06 0.44   3.00   3.04 0.44 2.0 4.4   2.4 0.31
## Petal.Length  3 150 3.76 1.77   4.35   3.76 1.85 1.0 6.9   5.9 -0.27
## Petal.Width   4 150 1.20 0.76   1.30   1.18 1.04 0.1 2.5   2.4 -0.10
## Species*      5 150 2.00 0.82   2.00   2.00 1.48 1.0 3.0   2.0 0.00
##          kurtosis   se
## Sepal.Length    -0.61 0.07
## Sepal.Width      0.14 0.04
## Petal.Length    -1.42 0.14
## Petal.Width     -1.36 0.06
## Species*        -1.52 0.07
# by group
psych::describeBy(iris, iris$Species)[1] # only first group
## $setosa
##          vars  n mean   sd median trimmed  mad min max range skew kurtosis
## Sepal.Length  1  50 5.01 0.35   5.0   5.00 0.30 4.3 5.8   1.5 0.11    -0.45
## Sepal.Width   2  50 3.43 0.38   3.4   3.42 0.37 2.3 4.4   2.1 0.04     0.60
## Petal.Length  3  50 1.46 0.17   1.5   1.46 0.15 1.0 1.9   0.9 0.10     0.65
## Petal.Width   4  50 0.25 0.11   0.2   0.24 0.00 0.1 0.6   0.5 1.18     1.26
## Species       5  50 1.00 0.00   1.0   1.00 0.00 1.0 1.0   0.0 NaN      NaN
##          se
## Sepal.Length 0.05
## Sepal.Width  0.05
## Petal.Length 0.02
## Petal.Width  0.01
## Species      0.00
```

Distribution of two categorical variables I

```
library(tidyr)
df_titanic <- as.data.frame(Titanic) |> tidyr::uncount(Freq)
# contingency table or crosstable
table_titanic <- table(df_titanic$Survived, df_titanic$Class)
table_titanic
##
##      1st 2nd 3rd Crew
##   No 122 167 528 673
##   Yes 203 118 178 212
# relative frequencies
prop.table(table_titanic)
##
##      1st      2nd      3rd      Crew
##   No 0.05542935 0.07587460 0.23989096 0.30577010
##   Yes 0.09223080 0.05361199 0.08087233 0.09631985
# marginal frequencies (1 = rows, 2 = cols)
margin.table(table_titanic, 1)
##
##   No  Yes
## 1490  711
```

Distribution of two categorical variables II

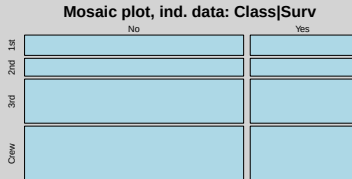
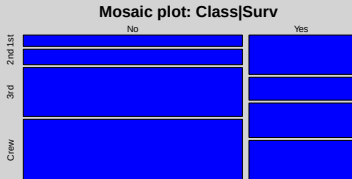
```
# conditional frequencies (1: rows, 2: cols)
prop.table(table_titanic, 2) # P(Survived|Class = column)
##
##           1st          2nd          3rd          Crew
##    No  0.3753846  0.5859649  0.7478754  0.7604520
##    Yes 0.6246154  0.4140351  0.2521246  0.2395480
par(mfrow = c(1, 3), bg = "lightgrey", mar = c(2, 2, 0.5, 0.5) )
# plot abs. frequencies
barplot(table_titanic, col = c("skyblue", "skyblue3"))
barplot(table_titanic, beside = TRUE, col = c("skyblue", "skyblue3"))
# plot cond. frequencies
barplot(prop.table(table_titanic, 2), beside = TRUE,
        col = c("skyblue", "skyblue3"))
```



Distribution of two categorical variables III

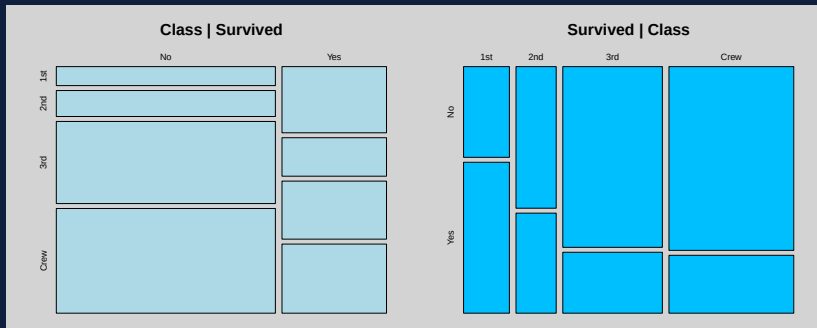
- ▶ Mosaic plot
 - ▶ **width** of bars: proportional to margin frequencies $P(X = x)$
 - ▶ **height** of bars: proportional to conditional frequencies $P(Y = y|X = x)$
 - ▶ **area**: proportional to joint frequencies $P(X = x, Y = y)$ (i.e., the cell count).

```
# create crosstable  $P(\text{Surv} = i, \text{Class} = j) = P(\text{Surv}=i)P(\text{Class} = j)$ 
etab <- outer(margin.table(table_titanic, 1),
              margin.table(table_titanic, 2))/sum(table_titanic)
par(mfrow = c(1, 2), bg = "lightgrey", mar = c(1, 2, 2, 0.5))
plot(table_titanic, main = "Mosaic plot: Class|Surv", col = "blue")
plot(as.table(etab), main = "Mosaic plot, ind. data: Class|Surv",
     col = "lightblue")
```



Distribution of two categorical variables IV

```
par(mfrow = c(1, 2), bg = "lightgrey", mar = c(1, 2, 3, 0.5))  
plot(table_titanic, main = "Class | Survived", col = "lightblue")  
plot(t(table_titanic), main = "Survived | Class", col = "deepskyblue")
```

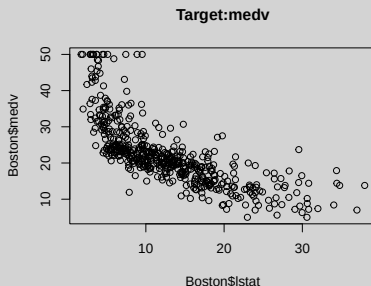
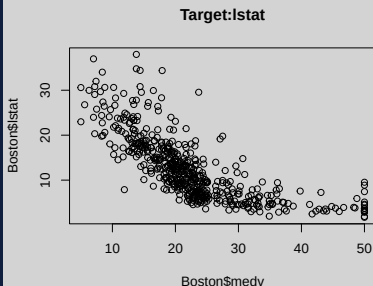


Mosaic plots are implemented in several R packages: `base mosaicplot()`, `vcd::mosaic()`, `ggmosaic::geom_mosaic()`.

Distribution of two continuous variables

- Scatter plot
 - y-axis: dependent/target variable
 - x-axis: independent variable/feature

```
par(mfrow = c(1, 2), bg = "lightgrey")  
library(MASS)  
plot(Boston$medv, Boston$lstat, main = "Target:lstat")  
plot(Boston$lstat, Boston$medv, main = "Target:medv")
```



Association measures (names, ranges, interpretation)

- ▶ Nominal variables
 - ▶ Chi-square statistic – range: ≥ 0
Larger values = stronger evidence against independence (size-dependent; not an effect size).
 - ▶ Contingency coefficient C – range: 0 to < 1
Larger = stronger nominal association (upper bound < 1 depends on table shape).
 - ▶ Corrected C* – range: 0 to 1
Larger = stronger nominal association (scaled version of C).
 - ▶ Cramer's V – range: 0 to 1
Larger = stronger nominal association (preferred effect size for $r \times c$ tables).
 - ▶ Cramer's Phi (2×2 only) – range: -1 to 1
Closer to 1 = strong positive association; closer to -1 = strong negative; near 0 = weak/none.
- ▶ Ordinal or metric variables
 - ▶ Spearman's rho – range: -1 to 1
Rank (monotonic) association: closer to 1 = strong positive; closer to -1 = strong negative; near 0 = weak/none.
 - ▶ Kendall's tau – range: -1 to 1
Rank association via concordant vs discordant pairs: interpretation as for rho.

Association between nominal variables

```
# chi square coeff (0 <= chi^2)
chisq.test(df_titanic$Survived, df_titanic$Class)$statistic
## X-squared
## 190.4011

# contingency coeff (0 <= cc < 1)
DescTools::ContCoef(df_titanic$Survived, df_titanic$Class)
## [1] 0.2821685

# corrected contingency coeff (0 <= ccc <= 1)
DescTools::ContCoef(df_titanic$Survived, df_titanic$Class,
                    correct = TRUE)
## [1] 0.3990465

# Cramer's V (0 <= V <= 1)
DescTools::CramerV(df_titanic$Survived, df_titanic$Class)
## [1] 0.2941201

# Cramer's Phi (only 2x2 tables)
DescTools::Phi(df_titanic$Survived, df_titanic$Sex)
## [1] 0.4556048
```

Correlation between ordinal or metric variables

► ordinal variables

```
# General Social Survey 2018
durl <- "https://gss.norc.org/Documents/stata/"
gss <- rio::import(paste0(durl, "2018_stata.zip"))
# kendalls tau (-1 <= tau <= +1) for trust in court and big
# business: 0 = No trust at all, ..., 10 = complete trust
cor(gss$trcourts, gss$trbigbus, use = "complete.obs",
    method = "kendall")
## [1] 0.4950149
```

► metric variables

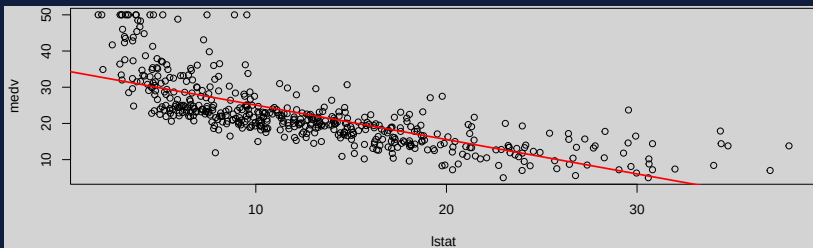
```
# Spearman's rank correlation (-1 <= rho <= +1)
library(MASS)
cor(Boston$lstat, Boston$medv, method = "spearman")
## [1] -0.8529141
# Pearson correlation (-1 <= r <= +1)
cor(Boston$lstat, Boston$medv)
## [1] -0.7376627
# Covariance
cov(Boston$lstat, Boston$medv)
## [1] -48.44754
```

Linear regression I

- ▶ Define a dependent variable Y and an independent variable X .
- ▶ Find a regression line $y = b_0 + b_1x$ such that the y_i are fitted well.
- ▶ To find the regression coefficients $\mathbf{b} = (b_0, b_1)^\top$ we use least squares.

$$\hat{\mathbf{b}} = \arg \min_{\mathbf{b}} \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

```
lreg <- lm(medv~lstat, data = Boston)
par(mar = c(4, 4, 0, 0) + 0.1, bg = "lightgrey")
plot(medv~lstat, data = Boston)
abline(lreg, col = "red", lwd = 2)
```



Linear regression II

```
lreg
##
## Call:
## lm(formula = medv ~ lstat, data = Boston)
##
## Coefficients:
## (Intercept)      lstat
##      34.55      -0.95
summary(lreg)
##
## Call:
## lm(formula = medv ~ lstat, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.168  -3.990  -1.318   2.034   24.500
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  34.55384    0.56263   61.41  <2e-16 ***
## lstat       -0.95005    0.03873  -24.53  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.216 on 504 degrees of freedom
## Multiple R-squared:  0.5441, Adjusted R-squared:  0.5432
## F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```


Linear regression III

- ▶ $b_0 = 34.55$
 - ▶ if %lower status people = 0 then $\hat{y} = 34.55$ (= 34550 US\$)
- ▶ $b_1 = -0.95$
 - ▶ if %lower status people increase by one percent then $\hat{y}$ will lower by 0.95 (= 950 US\$)
- ▶ $R^2 = 0.5441$
 - ▶ 54.41% of the variance in Y can be explained by the regression
 - ▶ 45.59% of the variance in Y can not be explained by the regression
- ▶ For classic time series analysis from Statistics I we will have an extra block

Common distributions I

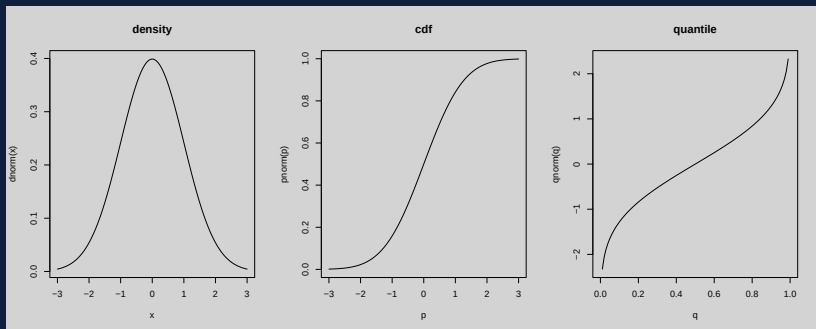
Distribution	Name	R function with parameter(s)
$B(n, p)$	Binomial	<code>dbinom(..., size = n, prob = p)</code>
$Hyp(N, M, n)$	Hypergeometric	<code>dhyper(..., m = M, n = $N - M$, k = n)</code>
$Po(\lambda)$	Poisson	<code>dpois(..., lambda = λ)</code>
$U(a, b)$	Uniform	<code>dunif(..., min = a, max = b)</code> ¹
$Exp(\lambda)$	Exponential	<code>dexp(..., rate = λ)</code> ²
$N(\mu, \sigma^2)$	Normal	<code>dnorm(..., mean = μ, sd = σ)</code> ³
χ_f^2	Chi square	<code>dchisq(..., df = f)</code>
t_f	t / Student	<code>dt(..., df = f)</code>
$F_{f1, f2}$	F / Snedecor	<code>df(..., df1 = $f1$, df2 = $f2$)</code>

► with

- `d` = density or probability function
 - `p` = cumulative distribution function
 - `q` = quantile function
 - `r` = random number generation
- Default for parameter(s): ¹ `min = 0`, `max = 1`, ² `rate = 1`, ³ `mean = 0`, `sd = 1`

Important distributions II

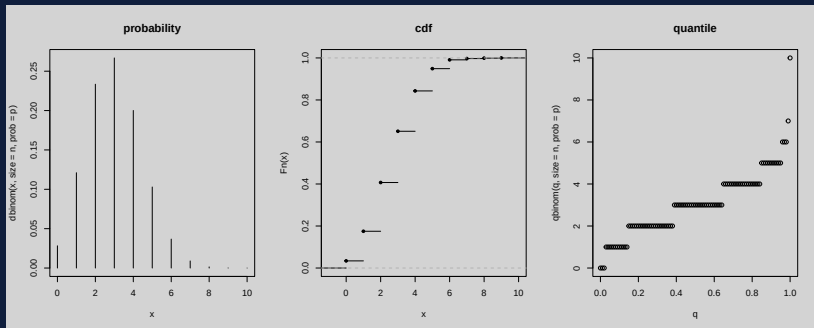
```
par(mfrow = c(1,3), bg = "lightgrey")
x <- p <- seq(-3, 3, length.out = 101) # N(0,1)
q <- seq(0, 1, length.out = 101) # cumulative probability as input
plot(x, dnorm(x), type = 'l', main = "density")
plot(p, pnorm(p), type = 'l', main = "cdf")
plot(q, qnorm(q), type = 'l', main = "quantile")
```



```
rnorm(5)
## [1] -1.68869233 -1.57237270 -0.40498716  0.31928642  0.04042768
```

Important distributions III

```
par(mfrow = c(1, 3), bg = "lightgrey")
n <- 10; p <- 0.3 # B(n = 10, p = 0.3)
x <- 0:n; q <- seq(0, 1, length.out = 101)
plot(x, dbinom(x, size = n, prob = p), type = 'h',
     main = "probability")
cdf <- ecdf(rbinom(1000, size = n, prob = p))
plot(cdf, main = "cdf")
plot(q, qbinom(q, size = n, prob = p), main = "quantile")
```



Sampling I

- ▶ **Simple random sampling (with replacement, SWR)**
 - ▶ Draw units with replacement from the population.
 - ▶ Standard assumption to draw conclusions about the population from a sample.
 - ▶ Sample variables X_i are independent and identically distributed (i.i.d).
- ▶ **Pure random sampling (without replacement, SOR)**
 - ▶ Draw units without replacement.
 - ▶ X_i are identically distributed but not independent.
 - ▶ Estimator variances are smaller than under SWR.
- ▶ **Survey statistics: other designs**
 - ▶ Some designs permit population inference.
 - ▶ Others do not.

Sampling II

```
index_swr <- sample(nrow(Boston), 100, replace = TRUE) # draw with replacement
table(index_swr)                                         # some obs. chosen more often

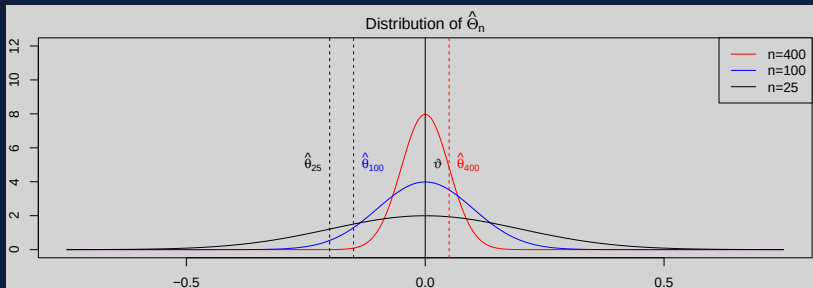
## index_swr
##   9  16  18  22  25  34  39  40  41  44  47  67  82  95 111 114 116 117 121 124
##   1   1   1   3   1   1   1   1   1   1   1   1   1   1   1   1   1   2   1   1
## 128 137 140 144 148 157 171 172 178 190 191 192 196 203 205 207 212 216 218 219
##   1   1   1   1   1   1   1   1   1   1   1   2   1   1   1   1   1   1   1   1
## 224 227 232 237 239 243 244 247 248 250 255 264 276 278 294 296 302 306 316 326
##   1   1   1   1   1   1   1   1   1   2   1   2   1   1   1   1   2   1   2
## 331 341 345 346 347 352 360 365 375 376 378 384 387 393 400 401 406 433 435 441
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 442 443 448 451 454 459 461 462 490 492 500 502
##   1   1   1   1   1   1   1   1   1   1   1   1

index_sor <- sample(nrow(Boston), 100)                  # draw without replacement
table(index_sor)                                         # each obs. chosen once

## index_sor
##   4   8  10  15  36  37  40  41  43  46  49  52  59  64  66  68  72  79  80  86
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
##  87  88  90  95  97 103 105 112 113 115 118 120 130 133 138 139 143 145 149 153
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 155 158 171 173 175 207 215 223 232 241 248 254 255 258 261 274 276 277 281 282
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 289 291 292 298 302 303 306 311 315 316 318 341 343 346 348 350 352 360 372 373
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 377 378 379 390 391 397 399 411 418 422 423 438 444 445 451 452 456 459 499 502
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
```

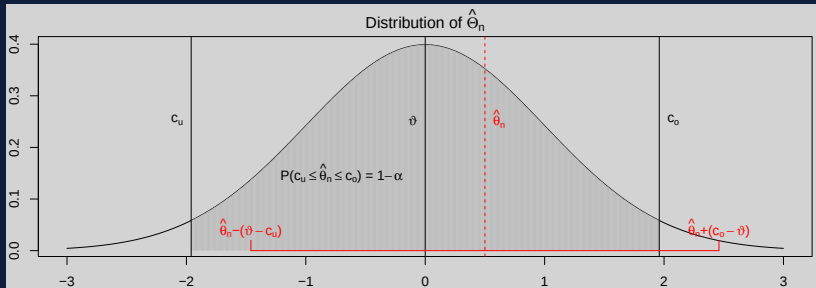
Estimators: methods and properties

- ▶ **Goal:** build an estimator $\hat{\Theta}_n = f(X_1, \dots, X_n)$ for a population parameter ϑ .
- ▶ **Methods to build estimators**
 - ▶ Maximum likelihood or least squares
- ▶ **Properties of estimators**
 - ▶ **Unbiasedness:** $\text{Bias}(\hat{\Theta}_n) = \mathbb{E}[\hat{\Theta}_n] - \vartheta = 0$.
 - ▶ **Efficiency:** $\text{Var}(\hat{\Theta}_n)$ is minimal.
 - ▶ **Consistency:** $\lim_{n \rightarrow \infty} (\text{Bias}^2(\hat{\Theta}_n) + \text{Var}(\hat{\Theta}_n)) = 0$.
- ▶ **Consistency (interpretation):** the distance between ϑ and an estimate $\hat{\theta}_n = f(x_1, \dots, x_n)$ goes to zero as $n \rightarrow \infty$.



Confidence intervals I

- ▶ Fluctuation interval (gray): the estimated value $\hat{\theta}_n$ lies with probability $1 - \alpha$ between c_u and c_o
- ▶ c_u and c_o cannot be computed since ϑ is unknown
- ▶ However, the interval widths $\vartheta - c_u$ and $c_o - \vartheta$ can be approximated thanks to the CLT
- ▶ If $c_u \leq \hat{\theta}_n \leq c_o$ then it holds $\hat{\theta}_n - (\vartheta - c_u) \leq \vartheta \leq \hat{\theta}_n + (c_o - \vartheta)$
- ▶ Confidence interval (red): with a growing sample size, it will cover the parameter ϑ with $1 - \alpha$ percent of confidence



Confidence intervals II

```
# exclude NAs from number of siblings
sibs <- na.omit(gss$sibs)
# confidence interval for the mean
DescTools::MeanCI(sibs, conf.level = 0.95) # 1-alpha=95% (default)
##      mean   lwr.ci   upr.ci
## 3.149808 3.072527 3.227089
# confidence interval for the female proportion
nfemale <- sum(gss$sex == 2)
n <- length(gss$sex)
DescTools::BinomCI(nfemale, n, conf.level = 0.95)
##           est   lwr.ci   upr.ci
## [1,] 0.5519591 0.5317759 0.5719726
```

- ▶ The package DescTools contains computation of further confidence intervals: BinomDiffCI, BinomRatioCI, MeanDiffCI, MedianCI, MultinomCI, PoissonCI, and VarCI.

Statistical tests

- ▶ For a two-sided statistical test in the fluctuation interval the parameter ϑ is replaced by the hypothetical value θ_0 which allows to compute $c_u(\theta_0)$ and $c_o(\theta_0)$
 - ▶ if $c_u(\theta_0) \leq \hat{\theta}_n \leq c_o(\theta_0)$ then null hypothesis **cannot** be rejected (type-II error possible)
 - ▶ if $\hat{\theta}_n < c_u(\theta_0)$ or $\hat{\theta}_n > c_o(\theta_0)$ then null hypothesis can be rejected (type-I error possible)
- ▶ There can be several pairs of hypotheses (two-sided, left-sided, right-sided) depending on the type of test
- ▶ A test statistics V can be derived from the estimate $\hat{\theta}_n$; V usually asymptotically follows a known distribution
- ▶ To simplify the test decision the p -value is used
 - ▶ if $p\text{-value} < \alpha$ (significance level) then the null hypothesis can be rejected
 - ▶ if $p\text{-value} \geq \alpha$ (significance level) then the null hypothesis can **not** be rejected
- ▶ **Note: Tests may have assumptions which needs to checked!**

Z-test/normal test/Gauß test

- ▶ Assumption: σ is **known**, $\mu = E(V)$, X_i either normal or central limit theorem (CLT) holds
- ▶ Hypotheses:
 - ▶ two-sided: $H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$
 - ▶ left-sided: $H_0 : \mu \geq \mu_0$ $H_1 : \mu < \mu_0$
 - ▶ right-sided: $H_0 : \mu \leq \mu_0$ $H_1 : \mu > \mu_0$
- ▶ In practice never used and therefore not in base R implemented

```
data(iq, package = "UsingR") # IQ should have mu_0=100 and sigma=15
# replace \mu with V= mean(IQ) = 101.35, test against \mu_0 = 100
# Z = (101.35 - 100)/(15/sqrt(100))= 0.9
# p_value = 2*pnorm(-abs(0.9)) = 0.3681203
compositions::Gauss.test(iq, mean = 100, sd = 15)
##
##  one sample Gauss-test
##
## data:  iq
## T = 101.35, mean = 100, sd = 15, p-value = 0.3681
## alternative hypothesis: two.sided
# 0.3681 > 0.05 => null hypothesis mu = 100 can not be rejected
```

Student's t -Test

- ▶ Assumption: σ is **unknown**, $\mu = E(V)$, X_i either normal or CLT holds
- ▶ Hypothesis:
 - ▶ two-sided: $H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$
 - ▶ left-sided: $H_0 : \mu \geq \mu_0$ $H_1 : \mu < \mu_0$
 - ▶ right-sided: $H_0 : \mu \leq \mu_0$ $H_1 : \mu > \mu_0$

```
# V \approx 3.1498, sd = 1.907603, n = 2343
# T_stat = (3.1498 - 3)/(1.907603/sqrt(2343)) = 3.801106
# p-value = 2*pt(-abs(3.801106), df = 2342) = 0.0001477279
t.test(gss$sibs, mu = 3)
##
## One Sample t-test
##
## data:  gss$sibs
## t = 3.8013, df = 2342, p-value = 0.0001476
## alternative hypothesis: true mean is not equal to 3
## 95 percent confidence interval:
##  3.072527 3.227089
## sample estimates:
## mean of x
##  3.149808
```

Two-sample t -Test/Welch test

- ▶ Assumption: $X_{i,1}$ and $X_{i,2}$ are independent and either normal or CLT holds
- ▶ Hypothesis: usually $\omega_0 = 0$
 - ▶ two-sided: $H_0 : \mu_1 - \mu_2 = \omega_0$ $H_1 : \mu_1 - \mu_2 \neq \omega_0$
 - ▶ left-sided: $H_0 : \mu_1 - \mu_2 \geq \omega_0$ $H_1 : \mu_1 - \mu_2 < \omega_0$
 - ▶ right-sided: $H_0 : \mu_1 - \mu_2 \leq \omega_0$ $H_1 : \mu_1 - \mu_2 > \omega_0$
- ▶ Further R parameters
 - ▶ `var.equal = FALSE` (Welch test)
 - ▶ for manual computation and formula of the test statistic, see [this website](#).

```
t.test(sibs~sex, data = gss)
##
##  Welch Two Sample t-test
##
## data:  sibs by sex
## t = -2.9062, df = 2246.9, p-value = 0.003694
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.38490454 -0.07474718
## sample estimates:
## mean in group 1 mean in group 2
##      3.022879      3.252705
# 0.01723 < 0.05 => null hypothesis can be rejected
```

Binomial test for a proportion π_0

- ▶ Assumption: X_i are binary
- ▶ Hypothesis:
 - ▶ two-sided: $H_0 : \pi = \pi_0$ $H_1 : \pi \neq \pi_0$
 - ▶ left-sided: $H_0 : \pi \geq \pi_0$ $H_1 : \pi < \pi_0$
 - ▶ right-sided: $H_0 : \pi \leq \pi_0$ $H_1 : \pi > \pi_0$
- ▶ Test does not use the normal approximation

```
nfemale <- sum(gss$sex == 2); n <- length(gss$sex)
# V = number of success (nfemale)
# p_greater <- pbinom(nfemale - 1, n, 0.5, lower.tail = FALSE)
# p_less <- pbinom(nfemale, n, 0.5, lower.tail = TRUE)
# 2 * min(p_less, p_greater) = 5.189887e-07
binom.test(nfemale, n, p = 0.5)
##
## Exact binomial test
##
## data:  nfemale and n
## number of successes = 1296, number of trials = 2348, p-value = 5.19e-07
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.5315772 0.5722112
## sample estimates:
## probability of success
##      0.5519591
# 5.19e-73 < 0.05 => null hypothesis can be rejected
# alternative test: prop.test, a chi^2 based test
```

Chi square goodness of fit test I

- ▶ Assumption:
 - ▶ all expected observations ≥ 5
- ▶ Hypothesis:
 - ▶ H_0 : X follows a discrete uniform distribution
 - ▶ H_1 : X follows not a discrete uniform distribution

```
# Is the dice fair? H_0: all numbers equally likely
# Test with n = 90 throws
expected <- c(15, 15, 15, 15, 15, 15) # H_0: 90 * 1/6 = 15
observed <- c(19, 13, 14, 12, 17, 15) # six faces
# V = sum((observed - expected)^2 / expected) = 2.266667
# pchisq(2.266, df = 5, lower.tail = FALSE) = 0.8112474
sum((observed - expected)^2 / expected)
## [1] 2.266667
chisq.test(observed, p = expected, rescale.p = TRUE)
##
## Chi-squared test for given probabilities
##
## data:  observed
## X-squared = 2.2667, df = 5, p-value = 0.8111
# 0.811 > 0.05 => null hypothesis can not be rejected
# no warning, thus assumption are fulfilled!
```

Chi square goodness of fit test II

- ▶ Assumption:
 - ▶ all expected observations ≥ 5
- ▶ Hypothesis:
 - ▶ H_0 : X follows a given distribution
 - ▶ H_1 : X follows not a given distribution
- ▶ Remarks: the distribution must be specified including the parameters

```
# H0: number customer/hour is Poisson dist., 50 hours observed
# 3h - 0, 6h - 1, 20h - 2, 17h - 3, ..., never more than 5 customers
customer <- c(0, 1, 2, 3, 4, 5, 6)      # 6 = 6 or more
# category labels for the hourly counts
observed <- c(3, 6, 20, 17, 7, 7, 0)
# how many hours in each category
lambda <- sum(customer*observed)/sum(observed); lambda
## [1] 2.666667
expected <- dpois(0:5, lambda = lambda) #P(X = 0), P(X = 1), ...
expected <- c(expected, 1 - sum(expected)) # 6 and more together
tst <- chisq.test(observed, p = expected) # assumption not fulfilled!
## Warning in chisq.test(observed, p = expected): Chi-squared approximation may
## incorrect
tst$expected # not expected <1, not many <5
## [1] 4.169007 11.117352 14.823136 13.176121 8.784081 4.684843 3.245460
```


Chi square goodness of fit test III

```
# approach 1: merge the first two and last two categories
observed5 <- c(observed[1] + observed[2], observed[3:6])
expected5 <- dpois(0:4, lambda = lambda)
expected5 <- c(expected5[1] + expected5[2], expected5[3:5], 1 - sum(expected5))
chisq.test(observed5, p = expected5)           # assumption now fulfilled!
##
## Chi-squared test for given probabilities
##
## data:  observed5
## X-squared = 5.9744, df = 4, p-value = 0.2011
# 0.2011 > 0.05 => the null hypothesis  $X \sim \text{Po}(2.6666)$  can not be rejected
# approach 2: estimate p-value by Monte Carlo simulation instead of chi-square
chisq.test(observed, p = expected, simulate.p.value = TRUE)
##
## Chi-squared test for given probabilities with simulated p-value (based
## on 2000 replicates)
##
## data:  observed
## X-squared = 10.353, df = NA, p-value = 0.1124
```

- ▶ the degrees of freedom should be reduced by the number of estimated parameters for the distribution (here: 1)
- ▶ therefore use other tests or make the number of categories large

Chi square independence test

- ▶ Assumption:
 - ▶ all expected observations ≥ 5
- ▶ Hypothesis:
 - ▶ H_0 : X and Y are independent
 - ▶ H_1 : X and Y are dependent

```
# gss$size = SIZE OF PLACE IN 1000S
# the + 3 shifts units by a factor of 1000 (since log10(1000) = 3)
lsize <- ceiling(3 + log10(gss$size))
# 3 = less than 1000 inhabitants, 4 = 1000-less than 10000, ...
lsize[lsize == -Inf] <- 3 # remove zeros: log10(0) = -Inf
ctab <- table(gss$sex, lsize);ctab
##      lsize
##      3   4   5   6   7
## 1  43 258 433 261  57
## 2  40 318 584 276  78
# expected in chi-squared E_ij = (row_total_i) (col_total_j)/N (rows: diff. sexes, col: diff. sizes)
# test stat V = sum((observed - expected)^2 / expected)
chisq.test(ctab)
##
## Pearson's Chi-squared test
##
## data:  ctab
## X-squared = 7.1855, df = 4, p-value = 0.1264
# 0.1264 > 0.05 => the null hypothesis of independence of sex and city size can not be rejected
```

Multiple linear regression I

- ▶ Linear regression $y_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p} + \varepsilon_i$
- ▶ Model with $Y_i = b_0 + b_1 x_{i,1} + \dots + b_p x_{i,p} + U_i$
- ▶ The least-squares estimator is BLUE (best linear unbiased estimator) under the following assumptions
 - ▶ $E(U_i) = 0$
 - ▶ $Var(U_i) = \sigma_U^2$ (homoscedasticity)
 - ▶ $Cov(U_i, U_j) = 0$
- ▶ On top of that
 - ▶ X_i are non-random (design matrix)
 - ▶ U_i describe only random effects, all systematic effects are captured by the X_i
 - ▶ between the X_i are no functional relationships (no multicollinearity)
- ▶ The regression function is linear in **parameters** (allows for $X_i^2, \dots$)
- ▶ For inference, we also $U_i \sim N(0, \sigma_U^2)$ or $U_i \sim (0, \sigma_U^2)$ and n large enough (such that CLT holds)

Multiple linear regression II

```
data(Boston, package = "MASS")
# medv: median house price in census district
# lstat: % lower status people in census district
# rm: average number of rooms in houses
lmbhd <- lm(medv~lstat + rm, data = Boston)
summary(lmbhd)
##
## Call:
## lm(formula = medv ~ lstat + rm, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.076  -3.516  -1.010   1.909   28.131
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.35827    3.17283  -0.428   0.669
## lstat       -0.64236    0.04373 -14.689 <2e-16 ***
## rm          5.09479    0.44447  11.463 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.54 on 503 degrees of freedom
## Multiple R-squared:  0.6386, Adjusted R-squared:  0.6371
## F-statistic: 444.3 on 2 and 503 DF,  p-value: < 2.2e-16
```

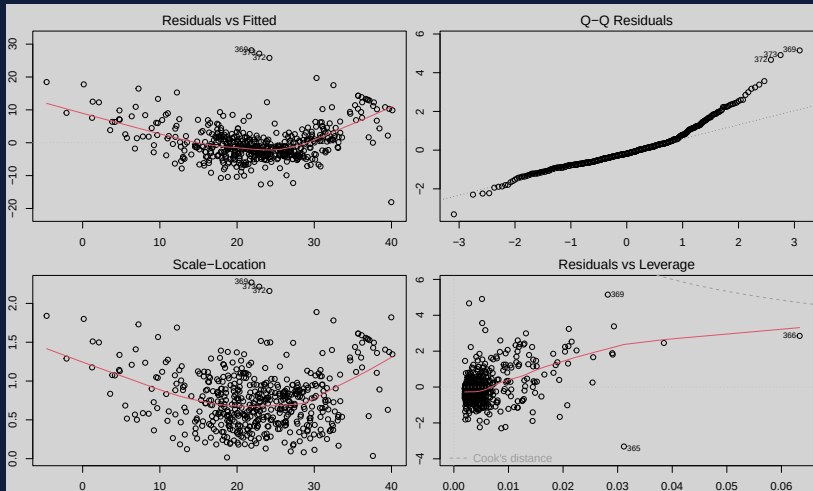
Multiple linear regression III

Interpretation of output:

- ▶ If residuals are normal/symmetric distributed then Median = 0 and $3Q \approx -1Q$ (not the case here)
- ▶ The estimated regression model is
$$\hat{y}_i = b_0 + b_1 lstat + b_2 rm = -1.35827 - 0.64236 lstat + 5.09479 rm$$
- ▶ The tests if the regression coefficients β_i are zero,
 - ▶ are rejected for β_1 and β_2 (p-value $< 2.2e-16$) and
 - ▶ not rejected for β_0 (p-value = 0.669)
- ▶ The $R^2 = 0.6384$ approx. 64% of the variance of Y are explained by the regression model
- ▶ Since $R^2 = 0.6384$ and $R^2_{adj} = 0.6371$ do not differ much all variables are necessary
- ▶ The test if $R^2 = 0$ or all $\beta_i = 0$ (for $i > 0$) is rejected
 - ▶ F-Test last line: p-value $< 2.2e-16$

Multiple linear regression IV

```
par(mfrow = c(2, 2), bg = "lightgrey", mar = c(2, 2, 2, 0.5)) # analys
plot(lmbhd) # and influential observations
```



Multiple linear regression V

► Useful functions

```
mylm <- lm(y~x)
# extract x and y used in estimating
mylm$model
# fitted values of the linear model
fitted(mylm); mylm$fitted
# residuals of the linear model
resid(mylm); mylm$resid
# coefficients, their variance
coef(mylm); mylm$coef
# variance covariance matrix of model
vcov(mylm)
# prediction
predict(mylm) # on current data
predict(mylm, newdata = newdf) # on new data
```

Multiple linear regression VI

Formula	Regression model
$y \sim x$	$y = \beta_0 + \beta_1 x$
$y \sim x + 1$	$y = \beta_0 + \beta_1 x$
$y \sim x - 1$	$y = \beta_1 x$
$y \sim x + z$	$y = \beta_0 + \beta_1 x + \beta_2 z$
$y \sim x + x : z$	$y = \beta_0 + \beta_1 x + \beta_2 x \cdot z$
$y \sim x * z - z$	
$y \sim x * z$	$y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 x \cdot z$
$y \sim (x + z + a)^3$	$y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 a + \beta_4 x \cdot z$ $\quad + \beta_5 x \cdot a + \beta_6 a \cdot z + \beta_7 x \cdot z \cdot a$
$y \sim I(x * z)$	$y = \beta_0 + \beta_1 x \cdot z$
$y \sim I(x^2)$	$y = \beta_0 + \beta_1 x^2$
$y \sim \text{poly}(x, 3)$	$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$